Piter Garcia
Color Histogram Descriptor

Update Report – Week 2

Work Done & In Progress

Formulating Algorithm - Image Representation:

In this section we describe the:

- color feature spaces
- texture feature spaces
- representation of distributions (in these spaces).

Color:

- the perceived differences between individual nearby colors
- implement the Euclidean distances between the color coordinates.

Texture: Color is a purely pointwise property of images;

- involves a notion of spatial extent, a single point has no texture.
- Frequency-domain texture descriptors refer instead to the frequency; content in a local neighborhood of the point.

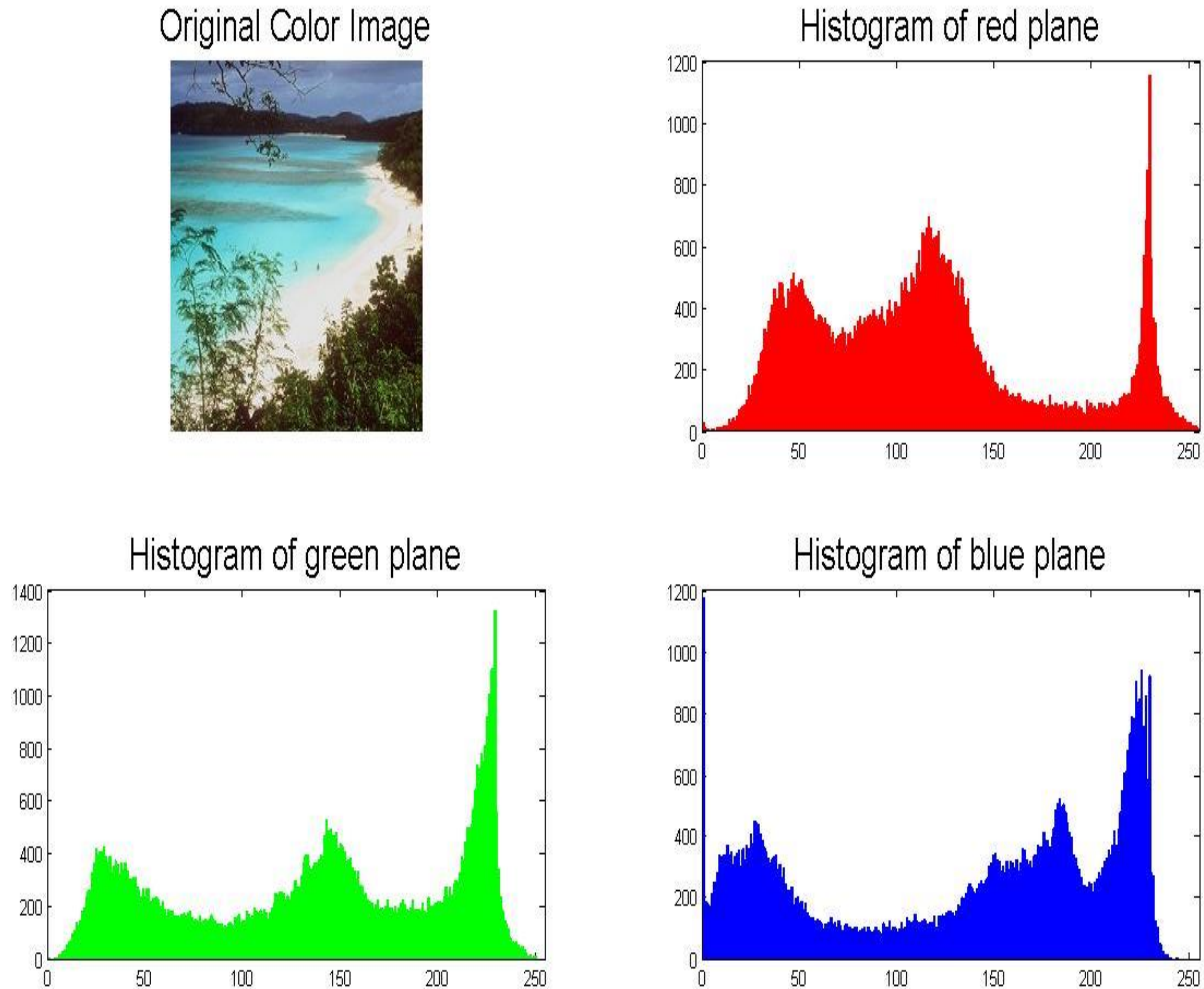
Work Done & In Progress

Obtaining and Analysing Bin Information

- feature selection
- image indexing
- retrieval
- pattern classification
- clustering.

Histogram Measure Methods:

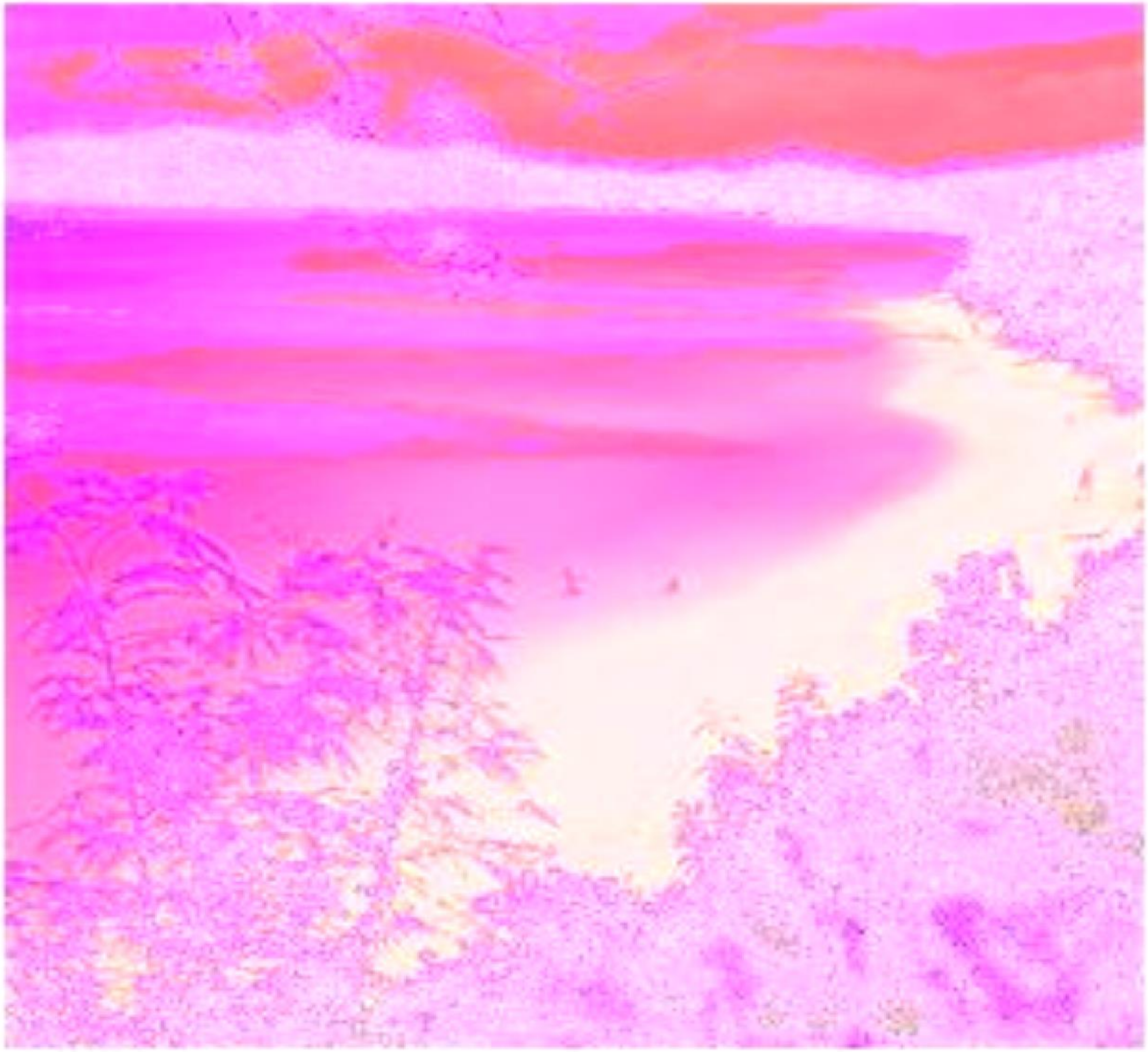
- Implementation & Comparison


Original Color Image
Histogram of red plane
0 200 400 600 800 1000 1200
0 50 100 150 200 250
Histogram of green plane
0 200 400 600 800 1000 1200 1400
0 50 100 150 200 250
Histogram of blue plane
0 200 400 600 800 1000 1200
0 50 100 150 200 250

Combining Histograms

- function newImg = rgb2lst(img)
- img = imread('OrangeFlower.jpg');
- newImg = uint8(zeros(size(img)));
- alpha = 1;

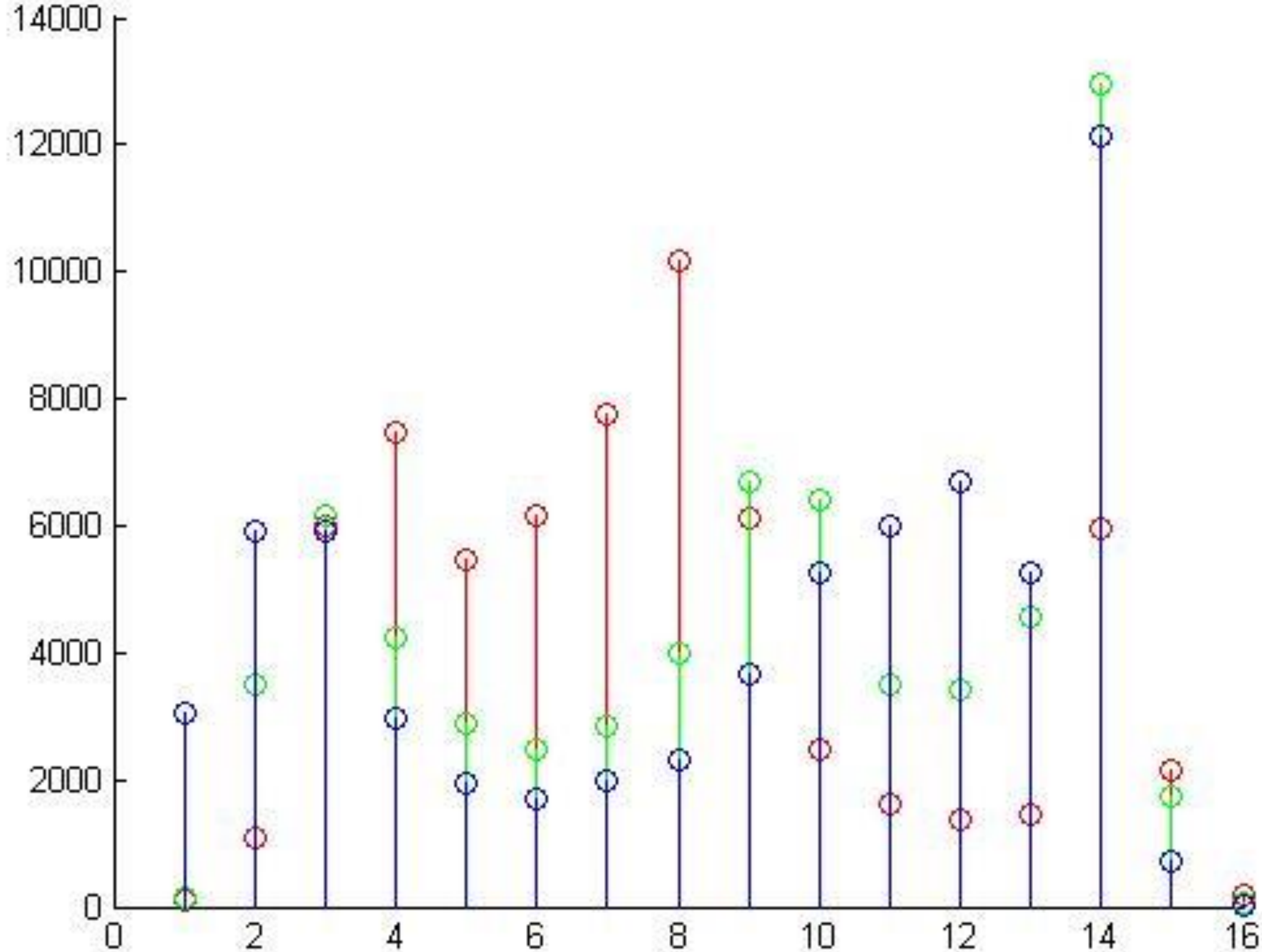
- lstMat =[alpha/sqrt(3), alpha/sqrt(3), alpha/sqrt(3); alpha/sqrt(2), 0,-alpha/sqrt(2); alpha/sqrt(6), 2*alpha/sqrt(6), alpha/sqrt(6)];
- %slow but steady
- for rowi = 1:(size(img,1))
-     for coli = 1:(size(img,2))
-      alpha = 255/max([img(rowi,coli,1),img(rowi,coli,2),img(rowi,coli,3)]);
-      lstMat =[alpha/sqrt(3), alpha/sqrt(3), alpha/sqrt(3);alpha/sqrt(2), 0, -alpha/sqrt(2); alpha/sqrt(6), 2*alpha/sqrt(6), alpha/sqrt(6)];
- newImg(rowi, coli,:) = uint8(double(lstMat)*double([img(rowi,coli,1);img(rowi,coli,2);img(rowi,coli,3)]));
-     end
- end
-



Combining Histograms

```
function threeDimHist = rgbhist(I,show)
%RGBHIST   Histogram of RGB values.

if (size(I, 3) ~= 3)
    error('rgbhist:numberOfSamples', 'Input image must be RGB.')
end

%change manually
nBins = 16;

rHist = imhist(I(:,:,1), nBins);
gHist = imhist(I(:,:,2), nBins);
bHist = imhist(I(:,:,3), nBins);

```

```
threeDimHist = cat(2,rHist, gHist);
threeDimHist = cat(2,threeDimHist, bHist);
%varargout = rHist;

if(show == true)
    figure;
    hold on
    stem(1:nBins, rHist,'-r');
    stem(1:nBins + 1/3, gHist,'-g');
    stem(1:nBins + 2/3, bHist,'-b');
    hold off
end;

```



14000
12000
10000
8000
6000
4000
2000
0
0
2
4
6
8
10
12
14
16

Bin Info & Comparison

threeDimHist =

102	157	3063
1093	3499	5928
6006	6171	5914
7456	4224	2955
5468	2874	1945
6179	2465	1695
7756	2864	1968
10188	3977	2315
6123	6686	3676
2458	6396	5245
1598	3511	6016
1351	3423	6691
1442	4549	5257
5968	12962	12143
2151	1721	724
197	57	1

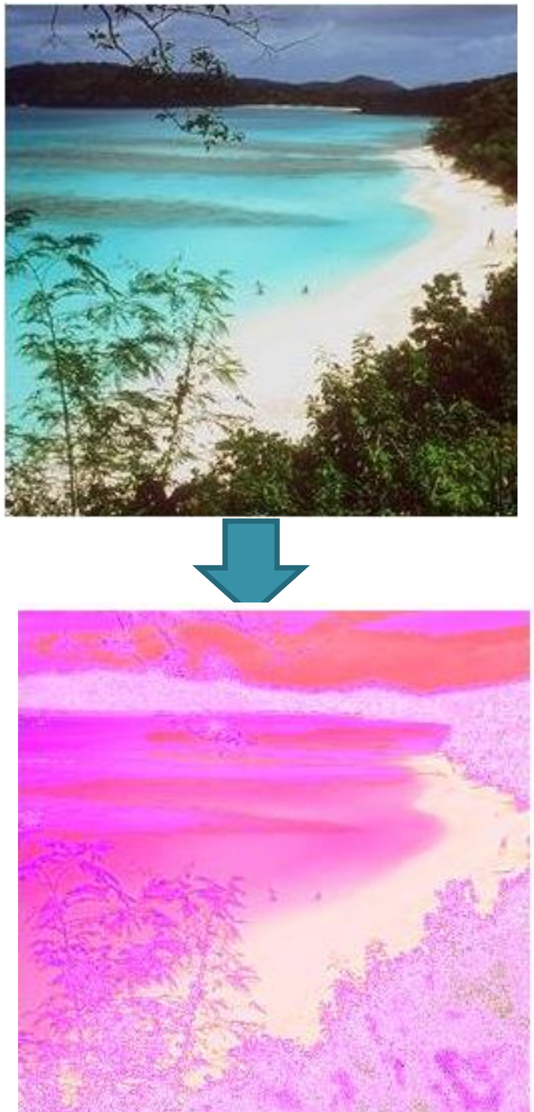
Testing Methods and Comparison

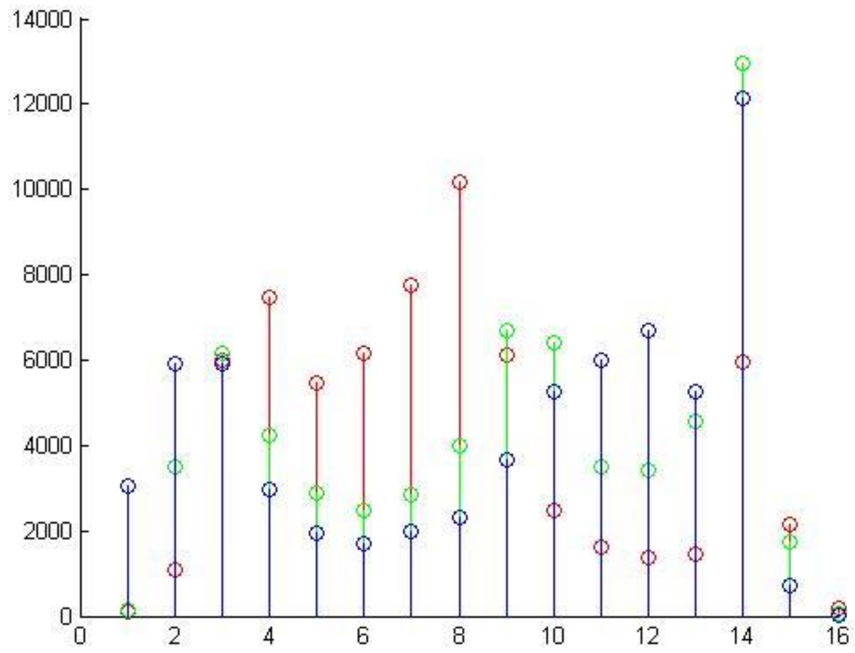
```
function d = CommonDist(X1,X2,dist_type)

switch dist_type
    case 'L1'
        d = sum(abs(X1-X2));
    case 'HI'
        d = sum(min(X1,X2));
    case 'Chi-square'
        d = sum(((X1-X2).^2)./((X1+X2)+1e-10));
    case 'BRD'
        tmp1 = abs(1-X1*X2');
        tmp2 = ((X1-X2).^2 +
         2*tmp1*(X1.*X2))./((X1+X2).^2+1e-10);
        d = abs(sum(tmp2));
```

Testing Methods and Comparison

```
 case 'L1-BRD'
        tmp1 = abs(1-X1*X2');
        tmp2 = ((X1-X2).^2 +
2*tmp1*(X1.*X2)).*abs(X1-X2)./((X1+X2).^2+1e-10);
        d = abs(sum(tmp2));
    case 'Chi-square-BRD'
        tmp1 = abs(1-X1*X2');
        tmp2 = ((X1-X2).^2 + 2*tmp1*(X1.*X2)).*(X1-
X2).^2./((X1+X2).^3+1e-10);
        d = abs(sum(tmp2));
    otherwise
        d = -1;
        error('unknown dist type!');
end
```

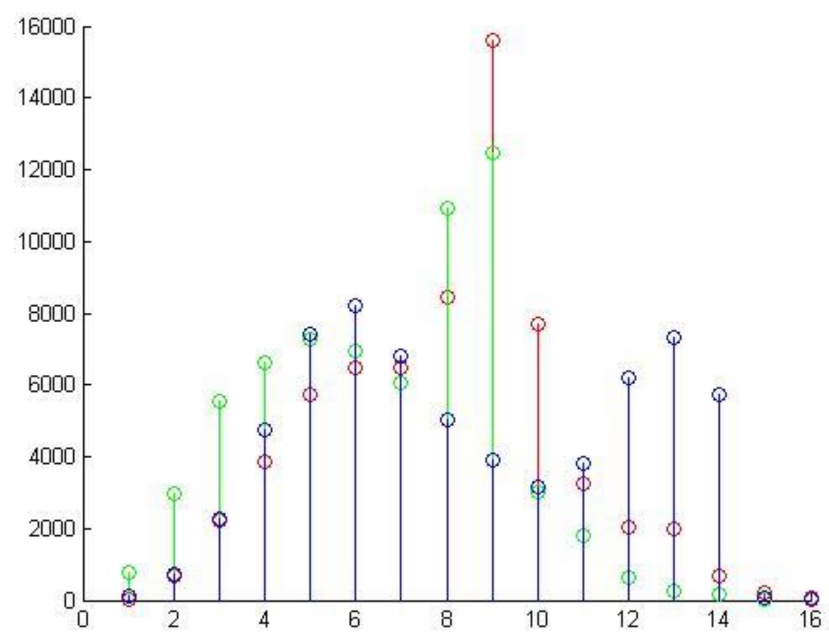
Comparing Coast & Tall-Building Classes





COAST COMBINED HISTOGRAM

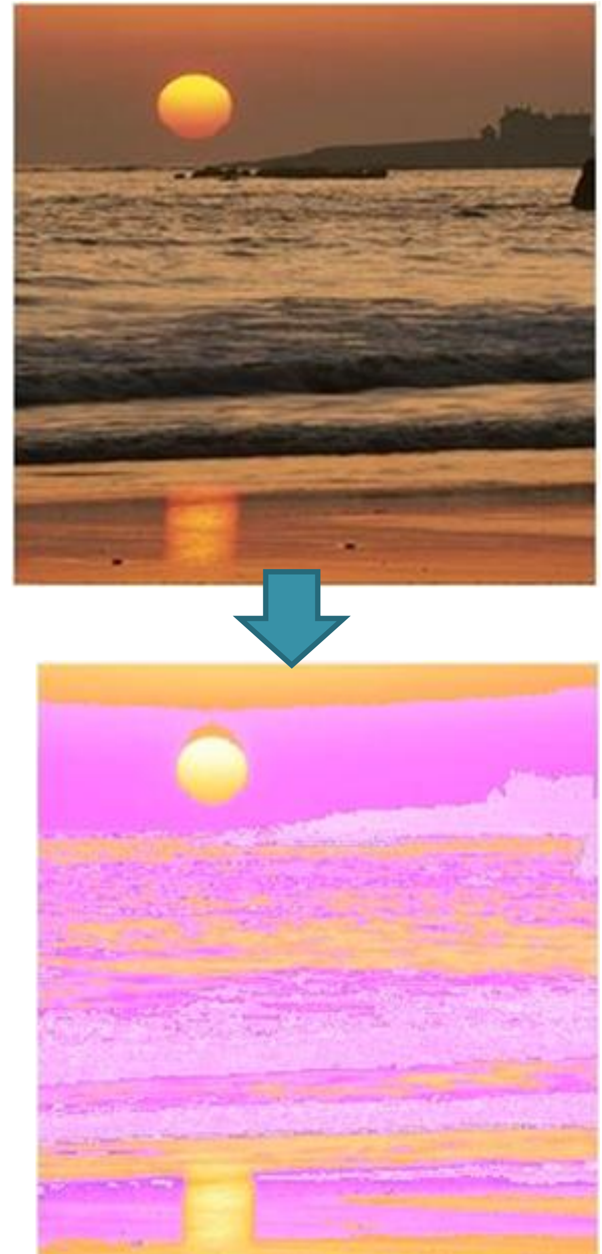
TALL BUILDING COMBINED HISTOGRAM

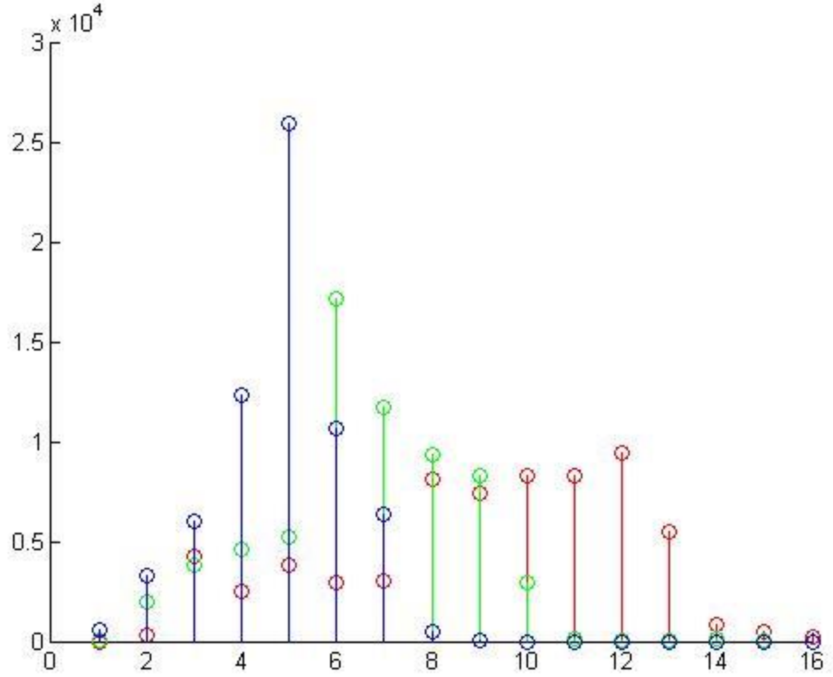


Comparison of Two Histograms of Different Classes

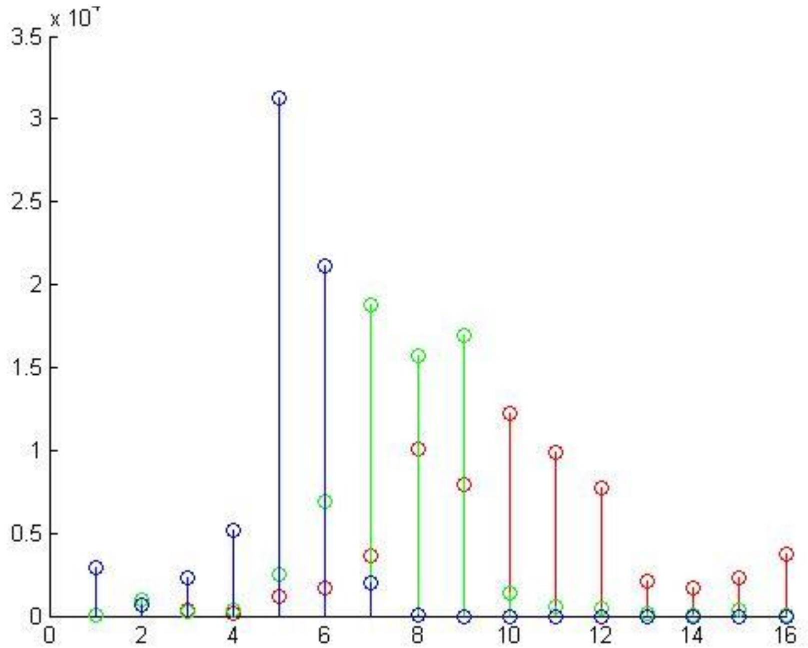
ans =			**L1**
36390	55684	47240	
ans =			**HI**
47341	37694	41916	
ans =			**BRD**
			1.0e+012 *
0.8162	0.3536	1.4950	
ans =			**L1-BRD**
			1.0e+014 *
1.1681	0.7481	0.5490	
ans =			**Chi-square**
			1.0e+004 *
1.6805	3.2903	2.4200	
ans =			**Chi-square-BRD**
			1.0e+013 *
6.1338	5.6304	4.1346	

Comparing Coast & Coast Class





COAST COMBINED HISTOGRAM



COAST COMBINED HISTOGRAM

Comparison of Two Histograms of the Same Class

ans = 30388	46934	36462	**L1**
ans = 50342	42069	47305	**HI**
ans = 0.0000	0.0000	1.1814	**BRD** 1.0e+014*
ans = 0.0166	0.0385	8.4154	**L1-BRD** 1.0e+014 *
ans = 1.4495	2.0363	1.4540	**Chi-square** 1.0e+004 *
ans = 0.0143	0.0236	8.4091	**Chi-square-BRD** 1.0e+016 *

Next On The List

- **Feature Extraction Further Development**

Individually, combine and average color histograms for each class

- **Set Comparison**

Create a list of category of the average histograms of each class

- **Apply Class & Class feature comparison**

Compare classes average histograms sets

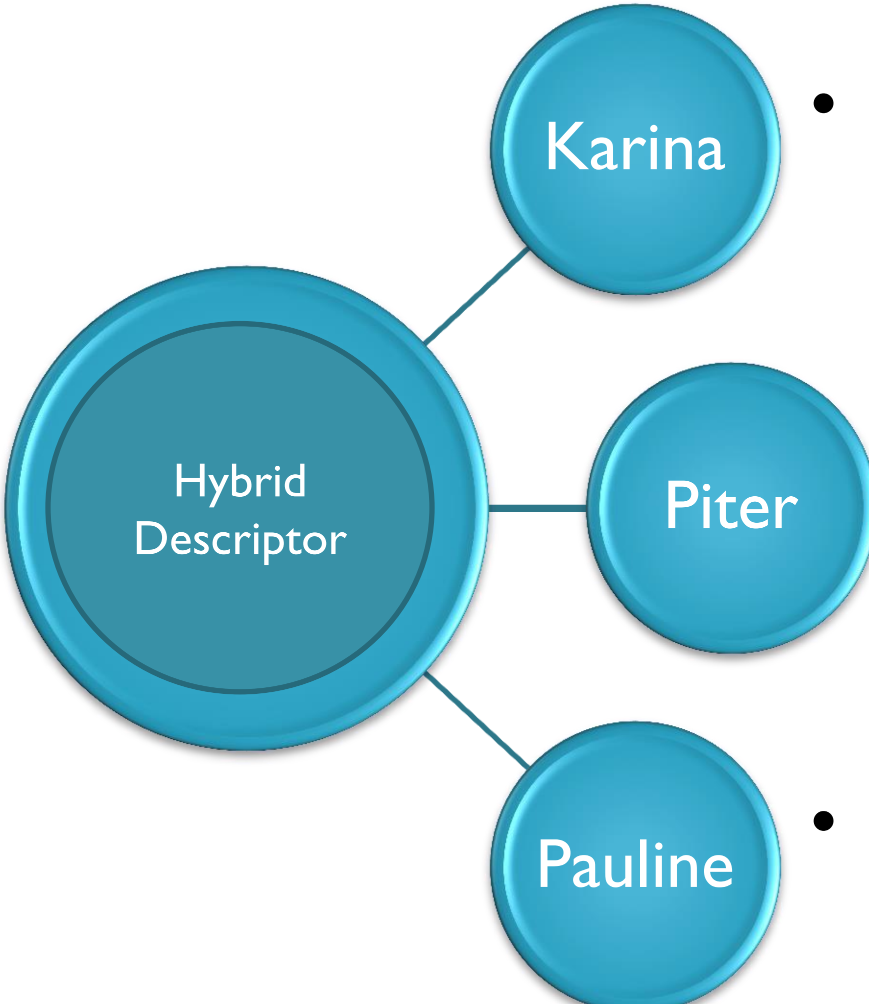
- Classification Comparison

Verify and separate data

- **Average Intersection   & Features Implementation**
- Check for intersecting features among classes

- Implement SVM

Project Flow Chart



- Gabor Descriptor
- Color Histograms Descriptor
- Centrist Descriptor

Some Benefits of a Hybrid Descriptor

Gabor - Adds better texture

SCENE CATEGORIES

Centrist – Adds better Visual and Object Recognition

Color Histograms – More Robust to Cluttering & Normalization